Worksheet 5A: Angles, Radians, and the Unit Circle

AIML Foundations Mathematics
Dublin and Dún Laoghaire ETB
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What This Worksheet Is About *Trigonometry began with ancient astronomers trying to measure the heavens. Today it's essential for everything from GPS satellites to semiconductor manufacturing. We start with two ways to measure angles (degrees and radians) and introduce the unit circle — a tool that will make everything else in trigonometry click into place.*

Part A: Degrees and Radians — Two Languages for Angles (20 problems)

Why Two Systems?

Degrees come from ancient Babylon (360° in a circle — probably because 360 has many divisors and roughly matches days in a year).

Radians come from mathematics itself: one radian is the angle where the arc length equals the radius.

$$\boxed{2\pi \text{ radians} = 360^\circ}$$

Therefore: π rad $= 180^\circ$, and 1 rad $= \frac{180^\circ}{\pi} \approx 57.3^\circ$

Converting Between Systems

Degrees → Radians: Multiply by $\frac{\pi}{180}$

Radians → Degrees: Multiply by $\frac{180}{\pi}$

Convert to radians (leave answers in terms of π):

1. $90^\circ =$ ______________________________ rad
2. $180^\circ =$ ______________________________ rad
3. $45^\circ =$ ______________________________ rad
4. $60^\circ =$ ______________________________ rad
5. $30^\circ =$ ______________________________ rad

6. $120° =$ ____________________________ rad
7. $270° =$ ____________________________ rad
8. $315° =$ ____________________________ rad
9. $150° =$ ____________________________ rad
10. $225° =$ ____________________________ rad **Convert to degrees:**
11. $\frac{\pi}{6}$ rad $=$ ____________________________$^\circ$
12. $\frac{\pi}{4}$ rad $=$ ____________________________$^\circ$
13. $\frac{\pi}{3}$ rad $=$ ____________________________$^\circ$
14. $\frac{2\pi}{3}$ rad $=$ ____________________________$^\circ$
15. $\frac{5\pi}{6}$ rad $=$ ____________________________$^\circ$
16. $\frac{5\pi}{4}$ rad $=$ ____________________________$^\circ$
17. $\frac{7\pi}{6}$ rad $=$ ____________________________$^\circ$
18. $\frac{11\pi}{6}$ rad $=$ ____________________________$^\circ$ **Science Connection: Why Radians Matter**
19. In calculus, the derivative of $\sin(x)$ is $\cos(x)$ — but **only if x is in radians!** If we used degrees, we'd get: $\frac{d}{dx}\sin(x^\circ) = \frac{\pi}{180}\cos(x^\circ)$ That ugly factor disappears with radians. This is why all scientific computing uses radians.
20. A satellite dish must be aimed within $0.1°$ of the correct angle. Convert this to radians.

Part B: Arc Length and the Radian Definition (12 problems)

The Key Formula

For a circle of radius r and central angle θ (in radians):

$$\boxed{s = r\theta}$$

where s is the **arc length**.

This formula is beautifully simple — but only works with radians!

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21. A circle has radius 10 cm. Find the arc length for a central angle of:
    (a) $\frac{\pi}{2}$ rad
    (b) π rad
    (c) 2π rad (verify this gives the circumference!)

22. Earth's radius is approximately 6,371 km. - (a) If you travel along Earth's surface through an angle of 1 radian (as measured from Earth's center), how far have you traveled? - (b) Dublin is at latitude 53.3°N. Convert this to radians. - (c) What is the arc length from the equator to Dublin along a meridian? **Science Connection: How Eratosthenes Measured Earth** Around 240 BCE, Eratosthenes noticed that at noon on the summer solstice, the sun was directly overhead in Syene (modern Aswan, Egypt) — it shone straight down a well. But in Alexandria, 800 km north, the sun cast a shadow at an angle of about 7.2° from vertical.

23. (a) Convert 7.2° to radians.

    (b) The arc length between the cities is 800 km. Using $s = r\theta$, solve for r (Earth's radius).

    (c) The actual radius is about 6,371 km. How close was this ancient measurement?

    Astronomy: Angular Size When we look at distant objects, we measure their **angular diameter** — how big they appear in the sky.

24. The Moon's angular diameter is about 0.52° as seen from Earth.

    (a) Convert to radians.

    (b) The Moon is about 384,400 km away. Using $s = r\theta$ (treating the Moon's diameter as an arc), estimate the Moon's actual diameter.

    (c) The actual diameter is about 3,474 km. How close is your estimate?

25. The Sun's angular diameter is also about 0.53° — almost the same as the Moon! This is why solar eclipses are possible. The Sun is about 150 million km away. Estimate the Sun's diameter.

26. Jupiter's angular diameter varies from 0.55 arcminutes to 0.83 arcminutes depending on its distance from Earth (1 arcminute = 1/60 of a degree).

    (a) Convert 0.83 arcminutes to radians.

    (b) At closest approach, Jupiter is about 588 million km away. Estimate its diameter.

Part C: The Unit Circle — Your New Best Friend (24 problems)

What Is It?

The **unit circle** is a circle with radius 1, centered at the origin.

For any angle θ measured from the positive x-axis: - The x-coordinate of the point on the circle is $\cos(\theta)$ - The y-coordinate is $\sin(\theta)$

$$\boxed{(\cos\theta, \sin\theta) = \text{coordinates on unit circle at angle } \theta}$$

Special Angles to Memorize

Angle (°)	Angle (rad)	$\cos\theta$	$\sin\theta$
$0°$	0	1	0
$30°$	$\frac{\pi}{6}$	$\frac{\sqrt{3}}{2}$	$\frac{1}{2}$
$45°$	$\frac{\pi}{4}$	$\frac{\sqrt{2}}{2}$	$\frac{\sqrt{2}}{2}$
$60°$	$\frac{\pi}{3}$	$\frac{1}{2}$	$\frac{\sqrt{3}}{2}$
$90°$	$\frac{\pi}{2}$	0	1

Memory trick: For sine at $0°$, $30°$, $45°$, $60°$, $90°$: $\frac{\sqrt{0}}{2}, \frac{\sqrt{1}}{2}, \frac{\sqrt{2}}{2}, \frac{\sqrt{3}}{2}, \frac{\sqrt{4}}{2}$

Find the exact values without a calculator:

27. $\cos(0) =$ ______________________ , $\sin(0) =$ ______________________
28. $\cos\left(\frac{\pi}{6}\right) =$ ______________________ , $\sin\left(\frac{\pi}{6}\right) =$ ______________________
29. $\cos\left(\frac{\pi}{4}\right) =$ ______________________ , $\sin\left(\frac{\pi}{4}\right) =$ ______________________
30. $\cos\left(\frac{\pi}{3}\right) =$ ______________________ , $\sin\left(\frac{\pi}{3}\right) =$ ______________________
31. $\cos\left(\frac{\pi}{2}\right) =$ ______________________ , $\sin\left(\frac{\pi}{2}\right) =$ ______________________

Extending to All Four Quadrants

ASTC Rule: Which functions are positive in each quadrant? - **A**ll (Quadrant I): Both sin and cos positive - **S**in (Quadrant II): Only sin positive - **T**an (Quadrant III): Only tan positive - **C**os (Quadrant IV): Only cos positive **Reference Angle:** The acute angle to the x-axis. Use symmetry! **Find the exact values:**

32. $\cos\left(\frac{2\pi}{3}\right) =$ ______________________ , $\sin\left(\frac{2\pi}{3}\right) =$ ______________________
Hint: $\frac{2\pi}{3}$ is in Q2, reference angle is $\frac{\pi}{3}$
33. $\cos\left(\frac{3\pi}{4}\right) =$ ______________________ , $\sin\left(\frac{3\pi}{4}\right) =$ ______________________
34. $\cos\left(\frac{5\pi}{6}\right) =$ ______________________ , $\sin\left(\frac{5\pi}{6}\right) =$ ______________________
35. $\cos(\pi) =$ ______________________ , $\sin(\pi) =$ ______________________
36. $\cos\left(\frac{7\pi}{6}\right) =$ ______________________ , $\sin\left(\frac{7\pi}{6}\right) =$ ______________________
37. $\cos\left(\frac{5\pi}{4}\right) =$ ______________________ , $\sin\left(\frac{5\pi}{4}\right) =$ ______________________
38. $\cos\left(\frac{4\pi}{3}\right) =$ ______________________ , $\sin\left(\frac{4\pi}{3}\right) =$ ______________________
39. $\cos\left(\frac{3\pi}{2}\right) =$ ______________________ , $\sin\left(\frac{3\pi}{2}\right) =$ ______________________
40. $\cos\left(\frac{5\pi}{3}\right) =$ ______________________ , $\sin\left(\frac{5\pi}{3}\right) =$ ______________________
41. $\cos\left(\frac{7\pi}{4}\right) =$ ______________________ , $\sin\left(\frac{7\pi}{4}\right) =$ ______________________

42. $\cos\left(\frac{11\pi}{6}\right) =$ ______________________ , $\sin\left(\frac{11\pi}{6}\right) =$ ______________________

Negative Angles and Angles Beyond 2π

43. $\cos(-\frac{\pi}{4}) =$ ______________________ , $\sin(-\frac{\pi}{4}) =$ ______________________
44. $\cos(-\frac{\pi}{6}) =$ ______________________ , $\sin(-\frac{\pi}{6}) =$ ______________________
45. $\cos\left(\frac{13\pi}{6}\right) =$ ______________________ , $\sin\left(\frac{13\pi}{6}\right) =$ ______________________
    Hint: $\frac{13\pi}{6} = 2\pi + \frac{\pi}{6}$
46. $\cos(3\pi) =$ ______________________ , $\sin(3\pi) =$ ______________________

The Pythagorean Identity

47. On the unit circle, a point is at $(\cos\theta, \sin\theta)$. The distance from this point to the origin is the radius, which equals 1. Using the distance formula: $\sqrt{\cos^2\theta + \sin^2\theta} = 1$ Square both sides: $\cos^2\theta + \sin^2\theta =$ ______________________ **This is the most important identity in trigonometry!**
48. Verify the identity for $\theta = \frac{\pi}{4}$: $\cos^2\left(\frac{\pi}{4}\right)+\sin^2\left(\frac{\pi}{4}\right) = \left(\frac{\sqrt{2}}{2}\right)^2+\left(\frac{\sqrt{2}}{2}\right)^2 =$ ______________________
49. Verify for $\theta = \frac{\pi}{3}$.
50. If $\sin\theta = \frac{3}{5}$ and θ is in Quadrant I, find $\cos\theta$.

Part D: The Tangent Function and More (14 problems)

Definition

$$\tan\theta = \frac{\sin\theta}{\cos\theta} = \frac{y}{x} = \text{slope of the line from origin to point on circle}$$

Find the exact values:

51. $\tan(0) =$ ______________________
52. $\tan\left(\frac{\pi}{6}\right) =$ ______________________
53. $\tan\left(\frac{\pi}{4}\right) =$ ______________________
54. $\tan\left(\frac{\pi}{3}\right) =$ ______________________
55. $\tan\left(\frac{\pi}{2}\right) =$ ______________________ (What happens here?)
56. $\tan\left(\frac{2\pi}{3}\right) =$ ______________________
57. $\tan\left(\frac{3\pi}{4}\right) =$ ______________________
58. $\tan(\pi) =$ ______________________ **Science Connection: Angles of Incidence and Reflection**

59. In optics, when light hits a surface, the angle of incidence equals the angle of reflection (measured from the perpendicular "normal" line). A laser beam hits a mirror at an angle of $\frac{\pi}{6}$ radians from the normal. - (a) What is the angle of reflection? - (b) What is the total deflection of the beam (angle between incoming and outgoing rays)?

60. In a periscope, light bounces off two mirrors. If each mirror is angled at 45° to the horizontal, through what total angle is the light path bent? **Microscopy: Numerical Aperture** The **numerical aperture** (NA) of a microscope objective determines its resolving power:

$$NA = n \cdot \sin(\theta)$$

where n is the refractive index of the medium and θ is the half-angle of the cone of light entering the objective.

61. An air-immersion objective ($n = 1.0$) has a half-angle of 60°.
    (a) Find $\sin(60°)$.
    (b) Calculate the numerical aperture.

62. An oil-immersion objective ($n = 1.52$) has a half-angle of 70°.
    (a) Find $\sin(70°)$ using a calculator (≈ 0.940).
    (b) Calculate the numerical aperture.
    (c) Why does using oil allow for higher NA and better resolution?

Lithography: The Diffraction Limit In semiconductor manufacturing, the smallest feature size that can be printed is limited by:

$$\text{Resolution} = k_1 \cdot \frac{\lambda}{NA}$$

where λ is the wavelength of light, NA is the numerical aperture, and k_1 is a process factor (typically around 0.25-0.4).

63. Extreme ultraviolet (EUV) lithography uses $\lambda = 13.5$ nm and has NA ≈ 0.33.
    (a) With $k_1 = 0.3$, what is the theoretical minimum feature size?
    (b) Current chips have 3nm features. Does this seem achievable with these parameters?

64. Older lithography used deep ultraviolet (DUV) light at $\lambda = 193$ nm with NA ≈ 1.35 (using immersion).
    (a) With $k_1 = 0.3$, what resolution is achievable?
    (b) Why did the industry need to move to EUV for smaller features?